L01: Introduction

S&DS 6165: Topics in Causal Inference

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Lecture outline

- Separate the causal target from choices made during design and analysis.
- Use a science table to define potential outcomes, SUTVA, and the finite-population average treatment effect.
- Define randomized designs and show how complete randomization identifies average potential outcomes and yields an unbiased estimator.

1 A course about choices

Causal inference begins with a comparison: what would happen under one course of action rather than another? Data do not answer that question on their own. We must decide what comparison matters, how treatment will be assigned, and how the resulting observations will be analyzed.

This course emphasizes **design-based inference**. We will treat the units and their potential outcomes as fixed and locate randomness in an assignment mechanism that we know because we chose it. That perspective keeps two kinds of choices visibly separate:

- **Design choices** determine the probability distribution used to assign treatment. Complete randomization, blocking, rerandomization, and clustered assignment are different designs.
- **Analysis choices** determine how the realized data are turned into an estimate. Difference in means, inverse-probability weighting, regression adjustment, and calibration are different analyses.

The causal target—the **estimand**—usually comes first and remains fixed while we compare designs and analyses. Sometimes the target itself changes. Trimming units with poor overlap in L04, for example, may improve the statistical problem by changing the population about which we ask the question. That is not merely a better estimator of the old target.

This distinction is a recurring discipline: **what is the question, what was randomized, and what was calculated?**

2 The comparison hidden in every unit

Consider six plots in an agricultural experiment. Label the plots $i = 1, \dots, 6$, and let the treatment label z equal 1 for fertilizer and 0 for control. For plot i, define $Y_i(z)$ to be the yield that would be observed if that plot were assigned treatment z. The plot therefore has two potential outcomes, $Y_i(1)$ and $Y_i(0)$, and their difference

$$\tau_i = Y_i(1) - Y_i(0)$$

is its **individual treatment effect**. Before discussing how treatment is assigned, imagine that both potential outcomes are written down for every plot.

Plot i	Yield under control $Y_i(0)$	Yield under treatment $Y_i(1)$	Individual effect $\tau_i = Y_i(1) - Y_i(0)$
1	10	13	3
2	12	12	0
3	8	12	4
4	15	14	-1
5	9	11	2
6	11	14	3

This is often called the **science table**. It is not a table we observe. Rather, it represents the fixed causal quantities that an experiment is meant to teach us about. To connect the table to observed data, we require a well-defined treatment, no interference between units, and consistency between assignment and observation; these requirements are stated more precisely below.

Let $Z_i \in \{0, 1\}$ denote plot i's actual assignment. We use an uppercase Z_i because, before randomization, the assignment is a random variable. Once the design draws its assignment, only

$$Y_i^{\text{obs}} = Y_i(Z_i) = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$$

is observed. If $Z_i = 1$, the formula selects $Y_i(1)$; if $Z_i = 0$, it selects $Y_i(0)$. The other entry in the row remains missing. This is the fundamental obstacle: the unit-level comparison is scientifically meaningful but cannot be observed jointly for the same unit.

The table also forces a substantive question. Why summarize the six effects by their average? The average effect is often a natural policy or welfare summary, and randomization makes it especially tractable. But the table may also contain harm, unequal gains, or effects concentrated in a subgroup. An average is a target, not a law of nature.

For this course, the default target is the **finite-population average treatment effect**

$$\tau = \frac{1}{N} \sum_{i=1}^{N} \tau_i = \overline{Y}(1) - \overline{Y}(0), \tag{1}$$

where N is the number of experimental units and

$$\overline{Y}(z) = \frac{1}{N}\sum_{i=1}^{N} Y_i(z)$$

is the finite-population mean under treatment level z. In the table, $\overline{Y}(1) = 76/6$, $\overline{Y}(0) = 65/6$, and therefore $\tau = 11/6$. These are features of these six plots; at this point no larger population or probability model for their yields has been invoked.

3 What the two-column table assumes

Writing $Y_i(z)$ assumes that the single label z contains all the treatment information needed to specify unit i's potential outcome. This bundles two ideas commonly called the **stable unit treatment value assumption** (SUTVA) (Rubin 1980):

1. **No multiple versions:** the treatment labeled z has no causally distinct, unrecorded versions. If several fertilizer formulations or application protocols matter for yield, the binary label must be refined or the versions must be declared irrelevant to the question.
2. **No interference:** unit i's outcome does not depend on other units' assignments. If fertilizer runs off from a neighboring plot, $Y_i(z)$ is generally not enough; the neighbors' assignments also matter.

Consistency then links the potential outcome to observation: if the realized assignment is $Z_i = z$, then $Y_i^{\text{obs}} = Y_i(z)$. SUTVA is not a metaphysical truth. It is a simplification whose credibility depends on how units and treatments are defined. L05 will replace $Y_i(z)$ with potential outcomes indexed by richer assignment or exposure patterns. The corresponding science table becomes much larger, but the logic—define the possible interventions before asking what is identified—survives.

4 Neyman's move: model the assignment, not the outcomes

One response to the missing entry is to model it. We might fit a linear model, posit a distribution for outcomes, or impute each unobserved potential outcome. Such strategies can be useful, but their conclusions depend on the outcome model.

The design-based alternative starts elsewhere. Hold the entire science table fixed and introduce a known random mechanism that reveals selected entries. The probability calculation then concerns which assignment vector was drawn, not how the potential outcomes were generated.

Fisher made randomization central to experimental inference, while Neyman's agricultural experiments and survey-sampling work developed the finite-population, design-based argument (Fisher 1935; Splawa-Neyman et al. 1990; Neyman 1934). In survey sampling, known inclusion probabilities let us learn about a fixed collection of units (Horvitz and Thompson 1952). In a randomized experiment, known treatment probabilities let us learn about a fixed collection of causal contrasts. Rubin's later formulation made the potential-outcomes language central across randomized and observational studies (Rubin 1974, 2005).

The contrast with an outcome model is therefore not "statistics versus no statistics." It is a choice about what supplies the probability statement:

design-based:	condition on the potential outcomes and average over $\mathbf{Z}$,
model-based:	posit a probability law for potential outcomes or observed responses, possibly conditional on treatment and pretreatment covariates.

In the first line, repeated-assignment statements concern the same plots and the same science table under different possible treatment assignments. In the second, probability also describes how outcomes vary across units or hypothetical samples. We will often combine the two—using regression, for example, to improve precision—but the assignment mechanism will remain the foundation for causal comparisons in randomized experiments.

5 A randomized design is a distribution

Let

$$\mathbf{Z} = (Z_1, \dots, Z_N) \in \{0, 1\}^N$$

be the assignment vector. Uppercase $\mathbf{Z}$ denotes the random vector before the design is realized; lowercase $\mathbf{z}$ denotes one particular assignment it could produce. A **randomized experimental design** is a probability distribution

$$\mathbb{P}_Z(\mathbf{Z} = \mathbf{z})$$

over possible assignment vectors $\mathbf{z}$. The subscript Z emphasizes that this probability comes from the assignment mechanism, with the potential outcomes held fixed. The support

$$\mathcal{Z} = \{\mathbf{z} : \mathbb{P}_Z(\mathbf{Z} = \mathbf{z}) > 0\}$$

records exactly which assignments the design can produce.

This definition matters. Saying "the study is randomized" is incomplete until we know both the support and the probabilities placed on it. Those choices determine which causal comparisons are possible and how estimators behave under repeated assignment.

5.1 Complete randomization

Under complete randomization, the experimenter fixes the treated count N_1 in advance and therefore also fixes the control count $N_0 = N - N_1$. Assume $0 < N_1 < N$, so both arms are nonempty. Every vector containing exactly N_1 ones is equally likely:

$$\mathbb{P}_Z(\mathbf{Z} = \mathbf{z}) = \begin{cases} \binom{N}{N_1}^{-1}, & \sum_{i=1}^N z_i = N_1, \\ 0, & \text{otherwise.} \end{cases}$$

There are $\binom{N}{N_1}$ ways to choose the treated units, which explains the probability in the first line. Each unit has marginal treatment probability N_1/N, but the assignments are not independent. If one unit is treated, one fewer treatment slot remains. In fact, for $i \neq j$,

$$\mathbb{E}_Z(Z_i Z_j) = \mathbb{P}_Z(Z_i = 1, Z_j = 1) = \frac{N_1(N_1-1)}{N(N-1)},$$

and therefore

$$\mathrm{Cov}_Z(Z_i, Z_j) = -\frac{(N_1/N)(N_0/N)}{N-1}.$$

That weak negative dependence is precisely what fixes the arm sizes.

5.2 Bernoulli randomization

Under an independent Bernoulli design with common probability p,

$$Z_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p), \qquad \mathbb{P}_Z(\mathbf{Z} = \mathbf{z}) = p^{\sum_i z_i}(1-p)^{N-\sum_i z_i}.$$

Now each unit receives an independent random draw, so the treated count $N_1 = \sum_i Z_i$ is random rather than fixed by the design. Throughout the course, N_1 and N_0 denote realized arm sizes: under complete randomization their values are fixed in advance, while under Bernoulli randomization they vary across possible assignments. The Bernoulli design is operationally simple and sometimes convenient for sequential enrollment. It can also produce imbalance. The probability of an empty treatment arm is $(1-p)^N$, and the probability of an empty control arm is p^N. We return to that complication in L02.

> **i** Design comes before positivity
>
> Write $\pi_i = \mathbb{P}_Z(Z_i = 1)$ for unit i's marginal treatment probability. **Positivity** is a property of the assignment mechanism relative to the causal comparison. For both arm-specific finite-population means to be learned without extrapolation, every unit's two relevant potential outcomes must each have a chance to be observed. Thus, for the treatment-control comparison,
>
> $$0 < \pi_i < 1 \qquad \text{for every } i.$$
>
> Under nondegenerate complete randomization, $\pi_i = N_1/N$; under the common-probability Bernoulli design, $\pi_i = p$. Notice that complete randomization satisfies this unit-level requirement even though it assigns probability zero to every vector with the wrong treated count. For more complicated exposures, the relevant requirement concerns the exposure contrasts defining the estimand—not an abstract rule that every assignment vector must be possible.

6 Why randomization identifies an average effect

Return to complete randomization. Here N_1 and N_0 are fixed, positive constants chosen by the design. The observed treated and control means are

$$\overline{Y}_1^{\text{obs}} = \frac{1}{N_1}\sum_{i=1}^{N} Z_i Y_i^{\text{obs}}, \qquad \overline{Y}_0^{\text{obs}} = \frac{1}{N_0}\sum_{i=1}^{N}(1-Z_i)Y_i^{\text{obs}}.$$

Consistency lets us rewrite these observable quantities in terms of the science table:

$$\overline{Y}_1^{\text{obs}} = \frac{1}{N_1}\sum_{i=1}^{N} Z_i Y_i(1), \qquad \overline{Y}_0^{\text{obs}} = \frac{1}{N_0}\sum_{i=1}^{N}(1-Z_i)Y_i(0).$$

For example, when $Z_i = 1$, the factor Z_i selects $Y_i(1)$; when $Z_i = 0$, that unit contributes zero to the treated total. Under complete randomization, $\mathbb{E}_Z(Z_i) = N_1/N$. The notation $\mathbb{E}_Z$ means that we average only over assignments produced by the design, holding all potential outcomes fixed. Because the arm sizes are fixed, their denominators can be taken outside the expectation:

$$\begin{aligned}
\mathbb{E}_Z\left(\overline{Y}_1^{\text{obs}}\right) &= \frac{1}{N_1}\sum_i \mathbb{E}_Z(Z_i)Y_i(1) = \overline{Y}(1), \\
\mathbb{E}_Z\left(\overline{Y}_0^{\text{obs}}\right) &= \frac{1}{N_0}\sum_i \mathbb{E}_Z(1-Z_i)Y_i(0) = \overline{Y}(0).
\end{aligned}$$

The cancellation is simple but important: each unit enters the treated average with probability N_1/N, and division by the fixed treated count turns those inclusion probabilities into equal weight $1/N$ on every $Y_i(1)$. The same argument applies to control.

These equalities are the design-based **identification** step: each fixed arm-specific mean is represented as the expectation of an observable random variable under the known assignment distribution. They do not identify either missing potential outcome for an individual unit. They also show why the distinction between complete and Bernoulli randomization matters. Under Bernoulli assignment, the denominators N_1 and N_0 are random and may equal zero, so one cannot repeat this calculation by simply moving them outside the expectation. L02 handles that case directly.

The corresponding **estimator** in the realized experiment is the difference in observed means,

$$\hat{\tau}_{\text{DM}} = \overline{Y}_1^{\text{obs}} - \overline{Y}_0^{\text{obs}}. \tag{2}$$

Linearity of design expectation then establishes its design unbiasedness:

$$\mathbb{E}_Z(\hat{\tau}_{\text{DM}}) = \tau. \tag{3}$$

The equality is a fact about the design. It does not require linearity, normal outcomes, constant effects, or a random sample from a larger population. It says that if we repeatedly reassigned these same N units while leaving their science table unchanged, the estimator would average to the fixed finite-population effect.

Randomization does not literally reveal both potential outcomes for any unit. It makes the observed treatment group representative of the finite population in expectation, and likewise for the control group. Identification concerns the average; unbiasedness concerns the repeated-assignment behavior of the estimator used to estimate it.

7 Estimand, estimator, estimate—and population

Three words prevent a great deal of confusion:

- The **estimand** is the fixed target, here τ in Equation 1.
- The **estimator** is the rule that maps the random assignment and resulting observed outcomes to $\hat{\tau}_{\mathrm{DM}}$. Before assignment, its value is random under the design.
- The **estimate** is the numerical value produced by the realized assignment and outcomes.

The target population must be stated just as carefully. Our τ averages over the N experimental units. It is a **finite-population** estimand. Even if those units were sampled from a larger population, treatment randomization by itself justifies the repeated-assignment statement for this realized collection of units.

We might instead posit a population distribution P for a generic unit's potential outcomes and target the **superpopulation** ATE

$$\tau_P = \mathbb{E}_P\{Y(1) - Y(0)\}.$$

Here $\mathbb{E}_P$ averages over units generated by P, rather than over alternative treatment assignments for the same units. Moving from the realized finite-population ATE to τ_P requires a sampling, transport, or superpopulation argument in addition to treatment randomization. A conditional average treatment effect,

$$\tau(x) = \mathbb{E}_P\{Y(1) - Y(0) \mid X = x\},$$

targets units with a particular value x of pretreatment covariates X. Under interference, one may instead define a global average treatment effect comparing two population-wide assignment regimes—for example, the average outcome if everyone were assigned treatment versus the average outcome if everyone were assigned control. These targets answer different questions; random assignment in one experimental sample does not automatically identify all of them.

8 What to carry forward

The science table states the causal comparisons. The design states which entries can be revealed and with what probabilities. The analysis states how the revealed entries are combined. Under complete randomization, the difference in observed means is design-unbiased for the finite-population average treatment effect. Every later refinement in the course should be locatable in that three-part structure.

9 Advanced material

9.1 Potential outcomes as missing data

Potential outcomes invite a missing-data reading: after assignment, one entry in each row of the science table is observed and the other is missing. More explicitly, the indicator that $Y_i(1)$ is observed is Z_i, while the indicator that $Y_i(0)$ is observed is $1 - Z_i$. Rubin's missing-data framework distinguishes mechanisms according to whether missingness depends on observed or unobserved values (Rubin 1976; Seaman et al. 2013). Very roughly:

- **MCAR**: missingness is independent of observed and missing values.
- **MAR**: conditional on observed information, missingness does not depend on the missing values.
- **MNAR**: dependence on missing values remains after conditioning.

Under complete randomization, assignment probabilities do not depend on the values in the fixed science table. In that limited, design-based sense, the missingness pattern for each treatment-specific outcome resembles MCAR: which entry is revealed is randomized independently of the values themselves. The assignment indicators remain dependent across units because the treated count is fixed, so the analogy should not be read as a claim that every missingness indicator is independent. More importantly, causal inference requires the interventions and their potential outcomes to be well defined; generic missing-data notation does not supply that scientific content.

9.2 Random arm sizes under Bernoulli assignment

Under i.i.d. Bernoulli assignment, $N_1 \sim \text{Binomial}(N, p)$. The usual difference in means is undefined when $N_1 = 0$ or $N_1 = N$. Conditional on any nondegenerate realized count $N_1 = n_1$, however,

$$\mathbb{P}_Z(\mathbf{Z} = \mathbf{z} \mid N_1 = n_1) = \binom{N}{n_1}^{-1} \qquad \text{when } \sum_i z_i = n_1.$$

The conditional design is therefore complete randomization with n_1 treated units, and the difference in means is conditionally design-unbiased. This does not make the ordinary difference in means unconditionally unbiased as a universally defined estimator: it still has no value on the two empty-arm events unless an extra convention is imposed. L02 introduces Horvitz–Thompson estimation, which uses the known assignment probabilities and remains defined and unbiased without conditioning on nondegeneracy.

9.3 Exact randomization variance under complete randomization

We now derive Neyman's exact variance formula rather than simply state it. The calculation has three parts: find the randomization variance of each observed arm mean, find their covariance, and then combine those pieces. The covariance matters because complete randomization selects complementary treatment and control groups rather than two independent samples.

The difference in means varies across assignments because different units enter each arm. To describe that variation, define the finite-population variances

$$S_z^2 = \frac{1}{N-1}\sum_{i=1}^{N}\{Y_i(z) - \overline{Y}(z)\}^2, \qquad S_\tau^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\tau_i - \tau)^2.$$

Here S_1^2 and S_0^2 are S_z^2 evaluated at $z=1$ and $z=0$. All three variances are fixed features of the science table. The only randomness below comes from the assignment.

9.3.1 Center the science table and recall the design moments

It will help to work with outcomes centered around their finite-population means. For this derivation only, write

$$a_i = Y_i(1) - \overline{Y}(1), \qquad b_i = Y_i(0) - \overline{Y}(0),$$

so that $\sum_i a_i = \sum_i b_i = 0$. Also define the finite-population covariance between the two columns of the science table,

$$S_{10} = \frac{1}{N-1}\sum_{i=1}^{N} a_i b_i.$$

Under complete randomization, exactly N_1 of the N units are treated. A specified unit is treated with probability N_1/N, so

$$\mathrm{Var}_Z(Z_i) = \frac{N_1}{N}\left(1 - \frac{N_1}{N}\right) = \frac{N_1 N_0}{N^2}.$$

For two distinct units i and j, the probability that both are treated is

$$\mathbb{E}_Z(Z_i Z_j) = \frac{N_1}{N}\frac{N_1 - 1}{N-1} = \frac{N_1(N_1-1)}{N(N-1)}.$$

Subtracting $\mathbb{E}_Z(Z_i)\mathbb{E}_Z(Z_j)$ gives

$$\mathrm{Cov}_Z(Z_i, Z_j) = -\frac{N_1 N_0}{N^2(N-1)}, \qquad i \neq j.$$

This negative covariance is the algebraic expression of sampling without replacement: once unit i occupies one of the fixed N_1 treatment slots, there is one fewer slot available to unit j. Treating the assignment indicators as independent would discard exactly the feature of complete randomization that produces the finite-population correction.

9.3.2 Variance of each arm mean

Centering the treated mean gives

$$\overline{Y}_1^{\text{obs}} - \overline{Y}(1) = \frac{1}{N_1}\sum_{i=1}^{N} Z_i a_i.$$

The finite-population mean $\overline{Y}(1)$ is fixed, so it does not affect the variance. Expanding the variance of the sum into its diagonal and off-diagonal pieces yields

$$\begin{aligned}
\operatorname{Var}_Z(\overline{Y}_1^{\text{obs}}) &= \frac{1}{N_1^2}\operatorname{Var}_Z\left(\sum_i Z_i a_i\right) \\
&= \frac{1}{N_1^2}\left\{\sum_i \operatorname{Var}_Z(Z_i)a_i^2 + \sum_{i\neq j}\operatorname{Cov}_Z(Z_i, Z_j)a_i a_j\right\} \\
&= \frac{1}{N_1^2}\left\{\frac{N_1 N_0}{N^2}\sum_i a_i^2 - \frac{N_1 N_0}{N^2(N-1)}\sum_{i\neq j} a_i a_j\right\}.
\end{aligned}$$

Here and below, $\sum_{i\neq j}$ runs over ordered pairs of distinct units. Thus both (i, j) and (j, i) appear.

The off-diagonal sum is not a new unknown. Because the a_i are centered,

$$0 = \left(\sum_i a_i\right)^2 = \sum_i a_i^2 + \sum_{i\neq j} a_i a_j,$$

and hence $\sum_{i\neq j} a_i a_j = -\sum_i a_i^2$. Substituting this identity, and using $\sum_i a_i^2 = (N-1)S_1^2$, gives

$$\begin{aligned}
\operatorname{Var}_Z(\overline{Y}_1^{\text{obs}}) &= \frac{1}{N_1^2}\frac{N_1 N_0}{N^2}\left(1 + \frac{1}{N-1}\right)\sum_i a_i^2 \\
&= \frac{N_0}{N N_1} S_1^2 = \left(\frac{1}{N_1} - \frac{1}{N}\right) S_1^2.
\end{aligned}$$

The factor $1/N_1 - 1/N$ is the usual sampling-without-replacement correction. By the same calculation, now treating the N_0 control units as a simple random sample from the fixed $Y_i(0)$ values,

$$\operatorname{Var}_Z(\overline{Y}_0^{\text{obs}}) = \left(\frac{1}{N_0} - \frac{1}{N}\right) S_0^2.$$

9.3.3 Covariance between the arm means

The two arm means are built from complementary groups, so their covariance must be included. Centering the control mean makes that dependence visible:

$$\begin{aligned}\overline{Y}_0^{\text{obs}} - \overline{Y}(0) &= \frac{1}{N_0}\sum_i (1 - Z_i) b_i \\ &= -\frac{1}{N_0}\sum_i Z_i b_i.\end{aligned}$$

The second equality uses $\sum_i b_i = 0$. Therefore,

$$\operatorname{Cov}_Z(\overline{Y}_1^{\text{obs}}, \overline{Y}_0^{\text{obs}}) = -\frac{1}{N_1 N_0}\operatorname{Cov}_Z\left(\sum_i Z_i a_i, \sum_j Z_j b_j\right).$$

Expand the covariance inside the last expression, again separating the terms with $i = j$ from those with $i \neq j$:

$$\operatorname{Cov}_Z\left(\sum_i Z_i a_i, \sum_j Z_j b_j\right) = \frac{N_1 N_0}{N^2}\sum_i a_i b_i - \frac{N_1 N_0}{N^2(N-1)}\sum_{i\neq j} a_i b_j.$$

Because both sets of outcomes are centered,

$$0 = \left(\sum_i a_i\right)\left(\sum_j b_j\right) = \sum_i a_i b_i + \sum_{i\neq j} a_i b_j.$$

Thus the off-diagonal sum is $-\sum_i a_i b_i$. Using the definition of S_{10} then gives

$$\operatorname{Cov}_Z\left(\sum_i Z_i a_i, \sum_j Z_j b_j\right) = \frac{N_1 N_0}{N} S_{10},$$

and consequently

$$\operatorname{Cov}_Z(\overline{Y}_1^{\text{obs}}, \overline{Y}_0^{\text{obs}}) = -\frac{S_{10}}{N}.$$

The minus sign comes from writing the centered control mean as a negative sum over treatment indicators. It does not imply that this covariance must be negative: its sign also depends on the finite-population covariance S_{10} between the two potential-outcome columns.

9.3.4 Assemble the pieces

Since $\hat{\tau}_{\mathrm{DM}} = \overline{Y}_1^{\mathrm{obs}} - \overline{Y}_0^{\mathrm{obs}}$,

$$\begin{aligned}\mathrm{Var}_Z(\hat{\tau}_{\mathrm{DM}}) &= \mathrm{Var}_Z(\overline{Y}_1^{\mathrm{obs}}) + \mathrm{Var}_Z(\overline{Y}_0^{\mathrm{obs}}) - 2\,\mathrm{Cov}_Z(\overline{Y}_1^{\mathrm{obs}}, \overline{Y}_0^{\mathrm{obs}}) \\ &= \left(\frac{1}{N_1} - \frac{1}{N}\right) S_1^2 + \left(\frac{1}{N_0} - \frac{1}{N}\right) S_0^2 + \frac{2S_{10}}{N}.\end{aligned}$$

The remaining step rewrites the covariance between the two columns in terms of treatment-effect heterogeneity. Since

$$\tau_i - \tau = \{Y_i(1) - \overline{Y}(1)\} - \{Y_i(0) - \overline{Y}(0)\} = a_i - b_i,$$

we have

$$\begin{aligned}S_\tau^2 &= \frac{1}{N-1}\sum_i (a_i - b_i)^2 \\ &= S_1^2 + S_0^2 - 2S_{10}.\end{aligned}$$

Substituting $2S_{10} = S_1^2 + S_0^2 - S_\tau^2$ into the preceding variance expression produces Neyman's exact formula (Splawa-Neyman et al. 1990):

$$\boxed{\mathrm{Var}_Z(\hat{\tau}_{\mathrm{DM}}) = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} - \frac{S_\tau^2}{N}}. \tag{4}$$

The first two terms are the arm-specific variance contributions before accounting for how the two potential outcomes are paired within units. The subtraction by S_τ^2/N records that pairing. If every unit has the same treatment effect, then $S_\tau^2 = 0$. With heterogeneous effects, $S_\tau^2 > 0$, so the exact variance is smaller than $S_1^2/N_1 + S_0^2/N_0$. This comparison describes the terms in the displayed decomposition; it does not say that changing a science table to create more heterogeneous effects must improve precision, since S_1^2 and S_0^2 may change as well.

A check using the six-plot science table. Suppose three of the six plots are assigned to treatment. From the complete table,

$$S_1^2 = \frac{22}{15}, \qquad S_0^2 = \frac{37}{6}, \qquad S_\tau^2 = \frac{113}{30}.$$

Neyman's formula therefore gives

$$\mathrm{Var}_Z(\hat{\tau}_{\mathrm{DM}}) = \frac{22/15}{3} + \frac{37/6}{3} - \frac{113/30}{6} = \frac{23}{12}.$$

There are $\binom{6}{3} = 20$ possible balanced assignments. Computing the difference in means under each one and averaging its squared deviation from $\tau = 11/6$ gives the same $23/12$.

The numerical agreement is a useful check on the algebra; the formula itself applies to any fixed science table under complete randomization.

We cannot generally identify S_τ^2 from the observed experiment. Its calculation requires knowing each pair $(Y_i(1), Y_i(0))$, yet randomization reveals only one member of each pair. Equivalently, the cross-column covariance S_{10} is unknown. By contrast, provided each arm contains at least two units, its observed outcomes let us estimate S_1^2 and S_0^2. Omitting the nonnegative S_τ^2/N term therefore gives an upper bound on the exact variance; replacing S_1^2 and S_0^2 with observed within-arm sample variances produces Neyman's conservative variance estimator, developed in L02.

10 Further reading

For the foundations of randomized experiments, read Fisher's *Design of Experiments* alongside Neyman's agricultural-experiment analysis (Fisher 1935; Splawa-Neyman et al. 1990). Neyman's representative-method essay and Horvitz–Thompson estimation make the survey-sampling lineage explicit (Neyman 1934; Horvitz and Thompson 1952). Rubin developed the modern potential-outcomes and missing-data presentation (Rubin 1974, 1976, 2005), while his comment on Fisher randomization connects potential outcomes to sharp-null randomization tests (Rubin 1980). Holland sharpened the distinction between causal effects and statistical association (Holland 1986).

Pearl's structural approach encodes causal claims through equations and graphs rather than beginning from a randomized assignment distribution (Pearl 1995, 2009). The two frameworks overlap in many applications, but this course develops the potential-outcomes, design-based route. For later examples of targets changing with the population or treatment received, see overlap-based trimming (Crump et al. 2009) and the local average treatment effect (Angrist et al. 1996).

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